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FOOD SCIENCE IN ACTION 2026

ABSTRACT BOOKLET

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FOOD SCIENCE IN ACTION 2026

5:00-5:15 PM	Check In
5:00-5:20 PM	Grab Dinner
5:25-5:30 PM	Opening Remarks
5:30-6:30 PM	Poster Presentations

The Effectiveness of a Laminated Antimicrobial Film to Control Microbes Associated with Low Moisture Foods

Matthew Shue[1] and Catherine Nettles Cutter[1]

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A laminated antimicrobial film (LAF) was made by coating an ethylene-vinyl alcohol (EVOH) film with an edible biopolymer layer that contains pullulan, water, gelatin, glycerol, and the antimicrobial, lauric arginate (LAE). In an earlier study, LAF effectively inhibited foodborne pathogens associated with fresh and ready-to-eat meat. This study explores the effectiveness of the LAF to control naturally occurring microbes associated with low moisture foods such as hard salami (a_w=0.8384) and aged Swiss (a_w= 0.9665). Foods were evaluated for initial microbial populations (aerobic plate counts [APC] and lactic acid bacteria [LAB]) upon receipt. Shortly thereafter, food was temperature abused at 25℃ for 12 hours to increase microbes and resulting microbial populations were determined on day 0. Low moisture foods were treated with one of three film types: a control film (EVOH only), LAF made without LAE, and LAF made with 0.5% LAE. Treated samples were stored at either 4℃ or 10℃ and evaluated for microbial populations at days 2, 7, 14, 21, and 28. Homogenized samples were serially diluted and plated onto respective media; incubated for 72 hours at 31°C (APC) and 48 hours at 31°C (LAB). The LAF with 0.5% LAE was most effective at reducing microbial populations in both hard salami and aged Swiss cheese over a 28-day period; reductions were less for hard salami than for cheese. This research highlights an important limitation for LAFs that rely on surface moisture to release their antimicrobial agent.

Spore than meets the eye: biochemical and mechanical differences in *Aspergillus oryzae* and *Neurospora intermedia* mycelium materials

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Fungal mycelium materials are popular for food, construction, and textile industries for their affordability. The fungal cell wall contains glucans and chitin, which are thought to contribute ductility and brittleness to mycelium materials, respectively. However, few studies link cell wall composition to mechanical properties, and growth conditions are not standardized. We hypothesize that a higher chitin:glucan ratio and higher protein content will correlate with mycelium mechanical strength. Standardized growth conditions were developed for mycelium materials from diverse fungal species. Chitin and glucans were measured using biochemical assays (modified Katano assay, Megazyme K-YGBL Yeast and Mushroom Kit™). Proteins, lipids, and total polysaccharides were measured using FTIR. Mycelium materials were tensile tested until failure in air. *A. oryzae* contained 37.0 ± 3.4% chitin and 21.8 ± 1.8% glucans, while *N. intermedia* contained 13.7 ± 6.1% chitin and 36.4 ± 6.0% glucans (chitin $p<0.001$; glucans $p=0.0014$; n = 6). *A. oryzae* had lower lipid and higher polysaccharide content than *N. intermedia* (lipids $p = 0.045$; polysaccharides $p = 0.026$; n = 3). Preliminary stress-strain testing suggested *N. intermedia* had significantly higher modulus at 10% strain (289.2 ± 74.0 kPa) compared to *A. oryzae* (111.3 ± 16.4 kPa; $p = 0.047$, n = 3). These findings suggest differences in water holding capacity, cell wall mechanics, or mycelium microstructure. Future studies will characterize mycelium microstructure and mechanical strength in wet conditions to understand the basis for these material differences. This research enables rational design of mycelium materials that support the biomanufacturing industry.

Improving Spirulina for Food Use: Effect of enzymatic treatment on composition and functionality

Danielle Stroinski [1], Yi Zhang[1]

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Spirulina is a cyanobacterium primarily used in the US market today as a nutritional supplement and as a natural food colorant. However, the cell composition of spirulina limits its solubility, which in turn reduces bioavailability. Spirulina Powder contains 60 – 80% protein, and a substaintial portion of its soluble proteins are phycocyanin, a protein-pigment complex giving spirulina its blue color. Other major components include peptidoglycan, a polymer composed of both sugar residues and peptide chains, as well as polysaccharides such as β-Glucans. These structural components can be degraded by exogenous enzymes; for example, lysozyme cleaves the β-1,4 glycosidic bond between NAM and NAG sugars, and β-glucanase hydrolyzes glycosidic bonds in glucans, promoting the generation of smaller polysaccharides and the release of spirulina proteins. Rather than focusing on the functionality of spirulina powder or spirullina protein concentrate, this study aimed to investigate the functional properties of enzymatically treated spirulina extract as novel food ingredients.

Influence of Hydrolysate Molecular Weight on the Formation and Emulsifying Properties of Pea Protein Fibrils

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In recent years, fibrillation has emerged as an efficient method to improve the functional properties of pea protein. However, the relationship between molecular weight distribution and the structural polymorphism of pea protein fibrils remains unclear. Thus, this research aimed to examine the effects of hydrolysate molecular weight on fibril formation. Moreover, the emulsifying properties of the fibrils were compared with their corresponding hydrolysates. Pea protein hydrolysates with various degrees of hydrolysis (DH2, DH6, DH10, and DH14) and their fibrillation products were analyzed using Transmission electron microscopy (TEM), thioflavin T fluorescence, intrinsic fluorescence, SDS-PAGE, and surface hydrophobicity. SDS-PAGE results indicated that not all large protein subunits participated in fibril formation, while higher degrees of hydrolysis (DH) values promoted the fibrillation process. TEM images showed that only DH14 formed long, straight fibrils, primarily composed of small peptides (10-20 kDa) that served as building blocks for fibril assembly. Furthermore, emulsion stability significantly improved after trypsin hydrolysis and fibrillation, with DH14 fibrils showing the highest emulsifying stability index, consistent with creaming index and visual appearance. Overall, this study deepens the understanding of subunit roles in amyloid fibrillation and expands the prospective applications of pea protein fibrils in the food industry.

Salivary proteins in bitterness perception in humans

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Bitterness is an aversive taste that can limit the acceptance of nutritionally beneficial foods like vegetables. Its perception begins with the dissolution of bitterants in saliva before they can activate taste receptors. Saliva is composed of water, ions, and critically, proteins, which may interact with bitterants in the oral cavity. Salivary proteins could influence taste by interacting with either taste receptors or chemical stimuli; to date, most evidence comes from animal studies, leaving the contribution of salivary proteins to bitterness perception in humans incompletely understood. Here, we explored binding interactions between human salivary proteins and quinine as a step toward understanding their potential role in bitterness perception. Saliva from 46 healthy volunteers was pooled and separated (3 kDa filter) into low and high molecular weight protein fractions. Protein-unbound quinine was quantified in whole and fractionated saliva using fluorescence spectroscopy, and binding was further characterized using isothermal titration calorimetry and analytical ultracentrifugation. The proportion of unbound quinine decreased with increasing protein concentration (to 59% at 0.6 mg/mL protein). Salivary proteins showed higher binding coefficient and reaction enthalpy, binding quinine comparably at 4.3X lower protein concentration than sodium caseinate, a food-grade protein previously studied by our team**.** Low molecular weight salivary proteins and peptides were particularly active at binding quinine, displaying the highest binding coefficient and enthalpy. This was further supported by the presence of protein-quinine complexes at higher sedimentation coefficients. This study demonstrates that human salivary proteins bind quinine, providing a foundation for future research into their role in bitterness perception.

Characterization of Distinct Sourdough Microbiomes and Their Impact on Bread Shelf Life Using Environmental and Challenge Studies

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The U.S. commercial baking industry typically relies on one yeast, *Saccharomyces cerevisiae*, for bread fermentation. In contrast, sourdough fermentation uses a complex community of naturally occurring bacteria and fungi. Shelf life is the time before bread becomes unacceptable in quality or safety due to textural changes (staling) or microbial growth. To extend bread shelf-life and prevent mold growth, commercial bakers use chemical preservatives. While effective, these additives conflict with the growing consumer demand for clean-label products. Sourdough fermentation can offer a solution by providing a natural way to decrease reliance on chemical preservatives and keep breads mold-free for longer. Previous work has demonstrated that microbial communities in sourdough produce organic acids and antimicrobial metabolites that can inhibit mold growth. Therefore, the objective of this study was to evaluate whether sourdough microbiomes can extend bread shelf-life comparable to or exceeding that of commercial preservatives. To accomplish this, we combined metagenomics with two experimental approaches: an environmental study and a challenge study. Twenty unique sourdough starters were propagated following standard laboratory procedures. Starters were mixed into doughs and baked into breads for shelf-life testing. Controls included unfermented dough, yeast-only fermentation, yeast with acidification to sourdough pH, and yeast with the chemical preservative calcium propionate (0.3%, equivalent to industry standards). The environmental study assessed mold spoilage native to the environment, while the challenge study tested the susceptibility of samples to common spoilage molds, *Aspergillus niger* and *Paecilomyces variottii*. We measured dough and bread pH and quantified organic acids. DNA was extracted from starters and subjected to shotgun metagenomic sequencing, followed by bioinformatic and statistical analyses. Sourdough breads exhibited significant differences in bread shelf-life performance. In the environmental study, ten sourdough breads outperformed the commercial bread standard with calcium propionate, and five showed no visible mold growth after 21 days. Data from the challenge study demonstrate that sourdough microbiomes varied in ability to inhibit the inoculated mold species. Organic acid analysis showed that acetic acid, but not lactic acid, was significantly correlated with increased bread shelf-life in sourdough samples (Spearman $\rho \approx 0.6$, $p < 0.05$). Preliminary taxonomic data suggests that no single species significantly correlates with shelf life. However, functional analysis identified genes, and metabolic pathways associated with antifungal activity present in sourdough microbiomes. Outcomes from this work demonstrate that sourdough fermentation can be used for industrial bread

production and reduce the need for synthetic preservatives while meeting consumer expectations for natural ingredients. Preliminary data further indicates that sourdough microbiome composition and functional may change over time, so future research will evaluate the stability of these communities. By leveraging microbiome structure and function, bread manufacturers can enhance product shelf life, minimize spoilage-related losses, and contribute to a more resilient food manufacturing ecosystem.

Persistence of *E. coli*, *Salmonella*, and *Listeria* from seeding to harvest in two soilless farming systems

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Limited data exist on the persistence of foodborne pathogens in soilless farming systems from seeding through harvest. The aim of this research was to characterize the persistence of *Salmonella, E. coli,* and *Listeria* in a soilless farming system over a full production cycle. We inoculated *Salmonella* Javiana and Thompson, *Escherichia coli* STEC O179:H8 and EPEC O13:H6, and two strains of *Listeria monocytogenes* at ~6 $\log_{10}$ CFU/mL in RO water and applied it to rockwool containing butterhead lettuce seeds (28 cubes per 3 system replicates). We assessed pathogen persistence during the 3-week nursery stage, as well as 4-week plant growth in deep water culture (DWC) or Kratky (KR) systems. We quantified pathogens in nutrient solution near plant roots and from the corners of each system throughout the growth stage, as well as in rockwool cubes and plant root surfaces at harvest. *E. coli* and *Listeria* concentrations declined in the nursery and plant growth stage ($p<0.001$). *Salmonella* counts increased during the nursery stage ($p=0.029$; 0.01 log units/day) but did not change during the plant growth stage in the systems. *E. coli* and *Salmonella* concentrations were lower at system corners compared to plant roots during the plant growth stage (linear mixed model, $p=0.047$, $p=7.14\times10^{-9}$, respectively). Pathogen concentrations did not differ between the DWC and KR systems. At the end of the growth period, all three pathogens were detected on roots (*Salmonella*: 3.1 ± 0.4, *E. coli:* 1.6 ± 0.6, *Listeria:* 0.6 ± 0.4 CFU/g) and rockwool (*Salmonella*: 4.6 ± 0.3, *E. coli:* 3.1 ± 0.9, *Listeria:* 0.6 ± 0.3 CFU/g). *Salmonella* concentration increased during the nursery stage and persisted through plant growth in the system. *E. coli* and *Salmonella* concentrations were higher near plant roots than at system corners, highlighting root-associated microenvironments as key hotspots for pathogen persistence independent of hydroponic system design.

Effects of Variety, Row Spacing, and Seeding Rate on Grain Nutritional Composition of Organic Spring Oats in Pennsylvania

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Oats are valued for their protein, fiber, and expanding role in gluten-free products. In organic production systems, agronomic management practices play an important role in maintaining grain quality. Row spacing and seeding rate can influence plant competition and stress responses, which may affect grain nutritional composition. Oats also contain ergothioneine (ERG), a sulfur-containing antioxidant produced primarily by soil microorganisms that has been associated with potential health benefits. Because ERG is microbially derived, its accumulation in oats may respond to agronomic conditions that influence plant stress and soil microbial activity. This study evaluated the effects of variety, row spacing, and seeding rate on oat grain nutritional composition and ERG concentration in organic spring oats grown in Pennsylvania. Three hull-less varieties (Gehl, Nitro, and Streaker) were grown under two row spacings (5.5 and 7.5 in.) and two seeding rates (1.5 and 2.5 million seeds acre^{-1}). Twenty-two nutritional parameters were quantified using near-infrared spectroscopy, and ERG concentrations were measured by high-performance liquid chromatography. Principal component analysis of nutrient traits revealed clear clustering by variety, indicating strong varietal differences in nutrient composition. ERG concentrations were significantly higher in Gehl under narrower row spacing (5.5 in.) compared to wider spacing (7.5 in.) (p = 0.019). Variety-specific trends were also observed between grain yield and crude protein, with Streaker and Gehl showing weak negative relationships while Nitro maintained relatively stable protein concentrations across yields. These results demonstrate that both genetic and agronomic factors shape the nutritional composition of organic oats and suggest that ERG accumulation may respond to management practices. Integrating variety selection with targeted agronomic strategies may provide opportunities to enhance the nutritional and functional value of organic oat production systems.

Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia

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In 2022, European Union member states reported 127 foodborne outbreaks attributed to *Bacillus cereus* toxins. Although foodborne illness is typically associated with *B. cereus sensu stricto* (*s.s.*) and biovar Emeticus, the recovery of biovar Thuringiensis from clinical cases raises concerns. The goal of this project was to characterize the genotypic and phenotypic diversity of 47 *B. cereus* group isolates recovered from human clinical cases by the Slovenian National Laboratory of Health, Environment, and Food between 2024-2025. DNA was extracted with the QIAGEN DNA Blood Mini Kit; libraries were prepared with the Rapid Barcoding Kit and sequenced on Nanopore MinION R9 flow cells. Genome assembly, polishing, and quality assessment were conducted with Flye (v2.9.5), Medaka (v1.3.2), and QUAST (v5.2.0). Taxonomic assignment and virulence gene detection were performed using BTyper3 (v3.4.0), while phylogenetic relationships were resolved using kSNP4 (v4.1.0). Maximum-likelihood phylogenetic trees were constructed with RAxML (Galaxy v8.2.12) and visualized in iTOL (v6). Cytotoxicity was assessed using Caco-2 cells exposed to 15% (v/v) culture supernatant, followed by WST-1 metabolic activity measurement. Isolates with good quality of assemblies (n=43) were classified as *B. cereus s.s.* (n=25, including 9 biovar Thuringiensis), *B. mosaicus* (n=12, including 1 biovar Emeticus), *B. mosaicus/luti* (n=2), and *B. toyonensis* (n=4). BTyper3 detected *nhe* genes in 86% of 43 isolates, *hbl* in 65%, *cytK-2* in 60%, and *sph* in 86%, while *cesABCD* was present in only one isolate (2%). Isolates from group IV (*B. cereus s.s.*) had the highest average cytotoxicity (0.57±0.32), whereas group II showed the lowest cytotoxicity (0.119±0.02), with no significant differences between *panC* groups (p=0.0716). *B. cereus s.s.* predominated among tested isolates and included 9 biovar Thuringiensis strains, 6 of which were closely related (>99% average nucleotide identity) to commercial biopesticide strains.

Distinct sourdough microbiomes alter FODMAPs of final breads

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FODMAPs (fermentable oligosaccharides, disaccharides, monosaccharides, and polyols) are poorly absorbed carbohydrates that can cause gut inflammation in individuals with inflammatory bowel diseases (IBDs). Commercial breads are believed to be high-FODMAP while sourdough breads may offer a low-FODMAP alternative that improves digestibility and nutritional quality.The objective of this study was to investigate the effect of twenty distinct sourdough microbiomes on bread FODMAP content. We hypothesized that fructan, the most dominant FODMAP in wheat, would be reduced in sourdough breads to below the clinical threshold of 0.3 g/serving compared to commercial yeast controls, and that total FODMAP content would vary depending on microbial composition. The significance of this work lies in identifying how microbial composition influences these outcomes. Twenty sourdough starters were obtained from Dr. Ben Wolfe (Tufts University) and propagated using a 1:1:1 starter-to-flour-to-water ratio. Breads were prepared under standardized laboratory conditions and analyzed for fructan and mannitol. Fructan was quantified using the Megazyme K-Fruc Assay, while mannitol was measured via HPLC. Shotgun metagenomic sequencing was performed for taxonomic and functional characterization. Distinct sourdough microbiomes produced significant differences in fructan and mannitol concentrations. Seven of twenty starters yielded breads below the low-FODMAP threshold; however, all sourdough breads contained higher fructan than the yeast control. Functional analysis identified microbial enzymes associated with FODMAP metabolism. These findings challenge the assumption that sourdough is inherently low-FODMAP and suggest that both microbial composition and fermentation conditions influence outcomes. Understanding these structure-function relationships will allow fermentation to be harnessed for developing clinically relevant low-FODMAP foods for individuals with IBDs.

Characterization of *Bacillus cereus* Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity Analysis

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The *Bacillus cereus* group is a diverse group of spore-forming bacteria classified into 8 *panC* phylogenetic groups, responsible for an estimated 63,000 cases of foodborne illness annually in the United States. *B. cereus* has been frequently detected at low levels in powdered infant formula and infant cereals. This study aimed to characterize *B. cereus* isolates from infant formula and cereal using genotypic and phenotypic methods. Isolates were recovered using the R&F® *Bacillus cereus* Group detection system for presumptive identification. Genomic DNA was extracted from presumptive *B. cereus* isolates and whole genome sequenced. Genome assembly, polishing, and quality assessment were performed through GalaxyTrakr using Flye (v2.9.5), Medaka (v1.3.2), and QUAST (v5.2.0), respectively. Taxonomic classification and virulence gene identification were performed with BTyper3 (v3.4.0), and phylogenetic grouping with kSNP4 (v4.1.0). RAxML (Galaxy v8.2.12) was used to build a maximum likelihood phylogenetic tree and visualized in the interactive tree of life (iTOL) (v6) software. Cytotoxicity was evaluated with human colorectal carcinoma (CaCo-2) cells. CaCo-2 cells were exposed to 15% (v/v) of treated supernatants, followed by the addition of WST-1 dye to measure metabolic activity through dye conversion. Isolates were identified as *B. mosaicus* (n=27), *B. cereus sensu stricto* (n=12), and *B. cytotoxicus* (n=2). High cytotoxicity was observed among several isolates across *panC* groups, with no significant differences detected between *panC* groups (p = 0.111), using a one-way ANOVA. These findings highlight the need for continued surveillance of *B. cereus* in infant foods to ensure the protection of vulnerable populations.

Effects of degree of hydrolysis on antioxidant activity of whey protein hydrolysates in nonionic stabilized oil-in-water emulsions.

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Lipid oxidation is a major cause of food spoilage, resulting in off-flavors and decreased nutritional value. Historically effective synthetic antioxidants have fallen out of favor due to potential toxicity. It is vital to find clean-label, safe, and cost-effective antioxidants to increase the shelf life of emulsions. This research introduced whey proteins into emulsions as antioxidants, allowing for the valorization of whey made during the cheese making process. Although many proteins possess inherent antioxidant capacity, their effectiveness can be increased through enzymatic hydrolysis. While previous studies have demonstrated the antioxidant potential of protein hydrolysates, their specific mechanisms of action in emulsions remain largely unexplored. We hypothesize that enzymatic hydrolysis of whey protein will reduce oxidation rates in emulsions, with smaller hydrolysates exhibiting the strongest antioxidant effects. Herein, whey protein isolate was hydrolyzed with degree of hydrolysis being controlled via enzyme to protein ratio or hydrolysis time. Radical scavenging and metal chelating abilities of hydrolysates were characterized by the Oxygen Radical Absorbance Capacity assay and the Ferrozine assay respectively. Results from these chemical assays were compared to the antioxidant effect of the hydrolysates in nonionic stabilized oil-in-water emulsions. Whey protein isolate and the hydrolysates were added in at concentrations ranging from 0.01 to 1 mg/g of emulsion. At low protein concentrations, it was shown that increased hydrolysis resulted in improved oxidative stability. The oxidation kinetics of this study were compared with a similar emulsion system in which micellar surfactant was removed, resulting in a much more rapid accumulation of lipid oxidation products. These results will aid the food industry in transitioning toward clean-label antioxidants while valorizing waste-stream proteins.